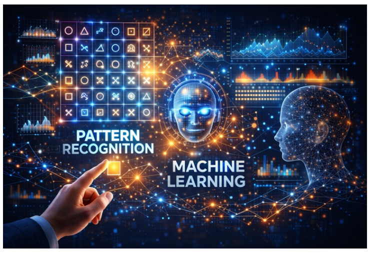
Introduction to Machine Learning
CSL2050 - Sem 02 – 2026
Lecture 24: Intro to Multi-Layer Perceptron

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IC: ChatGPT

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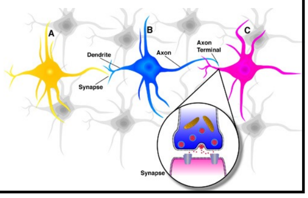
# Introduction to Neural Networks

## Motivation from Biology

- Brain is the key component in our body as it enables learning.
- It has a collection of about 10 billion interconnected **neurons**.
- A neuron receives input from other neurons (generally thousands) from its **synapses**
- Inputs are approximately summed
- Once the this sum exceeds a threshold the neuron sends an electrical spike that travels through the axon to the next neuron(s)

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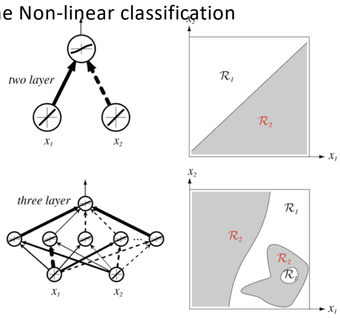
# Introduction to Neural Networks

- Natural Extension of LDF (Perceptron) to Multi Layer Perceptron for the Non-linear classification

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# Introduction to Neural Networks

- Natural Extension of LDF (Perceptron) for Non-linear classification
- Temporal History

| | |
|---|---|
| 1960 | • Hand Coded NN (No Learning) |
| 1985 | • Back Propagation NN (Chain Rule of GD) |
| 1990 | • Kernel Methods (Improved training) |
| 2000 | • Convolutional NN |
| 2010 | • Deep Learning (Scalable Learning) |

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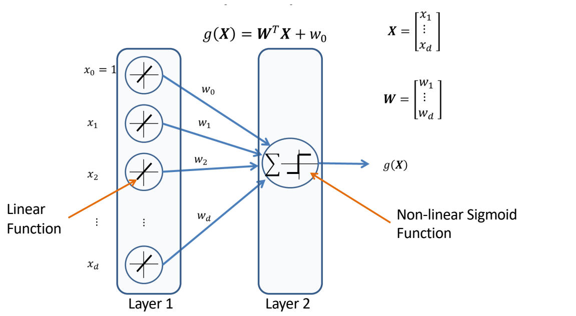
## Construction of Two Layer Perceptron

- Construction of a Two Layer Neural Network (Linear Classifier)

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## Construction of Multi Layer Perceptron

- Hidden Layer
- Bias Unit
- Neuron
- Net Activation
- Synapses/ Synaptic Weights
- Activation Function

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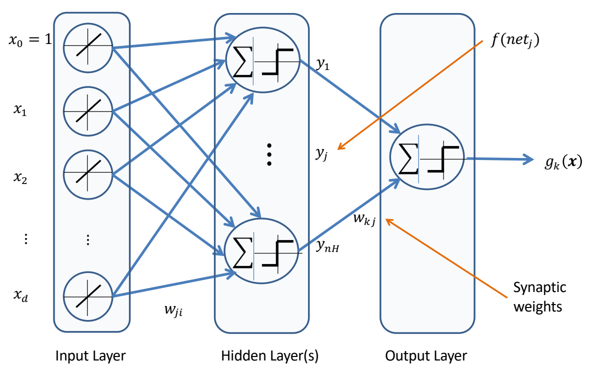
## Construction of MLP classifier

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## Feed Forward Operation in MLP

- $net_j = \sum_{i=1}^{d} x_i w_{ji} + w_{j0} = \sum_{i=0}^{d} x_i w_{ji} = \mathbf{w}^T{}_j\ \mathbf{x}$
- $y_j = f(net_j)$
- $y_j = sgn(net_j)$
- $net_k = \sum_{j=1}^{nH} y_j w_{kj} + w_{k0} = \sum_{j=0}^{nH} y_j w_{kj} = \mathbf{w}^T{}_k\ \boldsymbol{y}$
- $z_k = f(net_k) = sgn(net_k)$
- $g_k(\mathbf{x}) = z_k = \sum_{j=1}^{nH} w_{kj}\, f(\sum_{i=1}^{d} x_i w_{ji} + w_{j0}) + w_{k0}$

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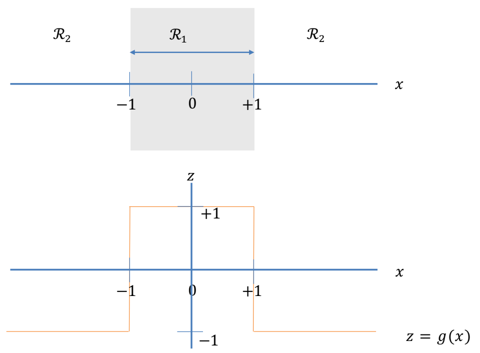
## Modelling the Non-linearity with MLP

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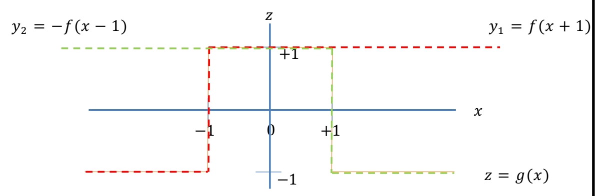
## Modelling the Non-linearity with MLP




$y_1 = sgn(w * x + b)$ $\quad$ $w = 1, b = +1$

$y_2 = sgn(w * x + b)$ $\quad$ $w = 1, b = -1$

$z = g(x) = sgn(w_1 y_1 + w_2 y_2 + b)$ $\quad$ $w_1 = +1, w_2 = -1, b = -0.5$

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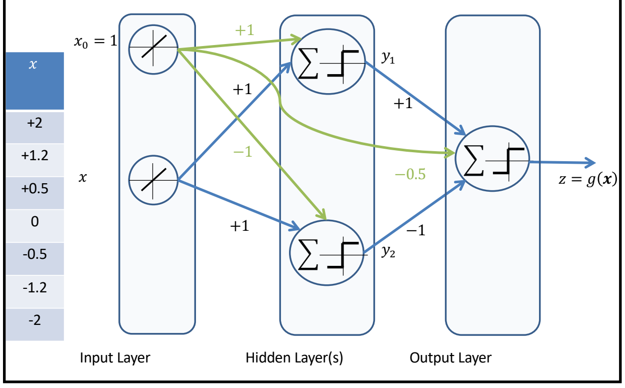
## Modelling the Non-linearity with MLP

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## Modelling the Non-linearity with MLP

| $x$ | $net_{j=1}$ | $net_{j=2}$ | $y_1$ | $y_2$ | $net_{k=1}$ | $z$ |
|---|---|---|---|---|---|---|
| +2 | +3.0 | +1.0 | +1 | +1 | -0.5 | -1 |
| +1.2 | +2.2 | +0.2 | +1 | +1 | -0.5 | -1 |
| +0.5 | +1.5 | -0.5 | +1 | -1 | +1.5 | +1 |
| 0 | +1.0 | -1.0 | +1 | -1 | +1.5 | +1 |
| -0.5 | +0.5 | -1.5 | +1 | -1 | +1.5 | +1 |
| -1.2 | -0.2 | -2.2 | -1 | -1 | -0.5 | -1 |
| -2 | -1.0 | -3.0 | -1 | -1 | -0.5 | -1 |

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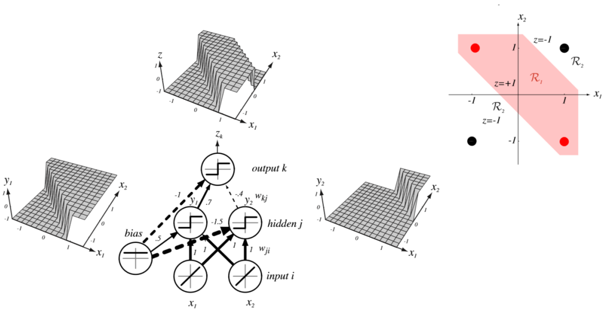
## Modelling the Non-linearity with MLP

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## Thank You

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